

Hybrid Robust Collaborative Perception with LiDAR-4D Radar Fusion under Adverse Weather Conditions

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Abstract

Current collaborative perception systems have significantly improved 3D object detection performance. However, widely used LiDAR and camera systems often suffer performance degradation under adverse weather conditions. The weather-robust 4D radar provides a promising solution to address this challenge. Nevertheless, the effective fusion of sparse 4D radar measurements with degraded LiDAR data remains a significant challenge due to cross-modal corruption and information loss. In this work, we propose a novel hybrid robust collaborative perception framework (HRCP), designed to improve the collaborative perception performance under adverse weather conditions through LiDAR-4D radar fusion. Specifically, we introduce a hybrid collaboration strategy that considers their distinct physical properties and differently processes them during information transmission. Additionally, we propose a bidirectional cross-modal gating (BCMG) module that enables LiDAR and 4D radar to mutually validate feature reliability, ensuring consistent cross-modal representation, and an adaptive feature enhancement (AFE) module that enables comprehensive refinement of degraded and suppressed regions to mitigate information loss. Extensive experimental results demonstrate that our method outperforms previous state-of-the-art approaches under adverse weather conditions.

1. Introduction

Perceiving complex driving environments is crucial to ensure the driving safety of agents (e.g., autonomous vehicles). With advancements in localization and perception technologies, even single-agent perception systems have demonstrated impressive performance in 3D object detection tasks [18, 42, 45]. Nevertheless, such systems suffer from inherent limitations, including occlusions and reduced detection accuracy for distant object instances. Collaborative perception has gained increasing attention as a promis-

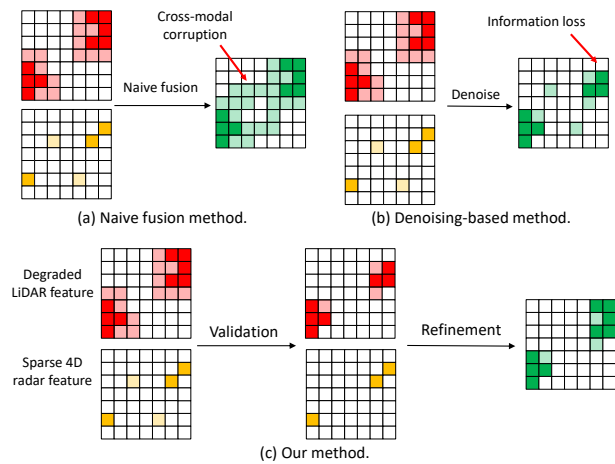


Figure 1. Different fusion methods. The red, orange, green regions are detected and located within the ground-truth bounding boxes from LiDAR, 4D radar and the fused feature, respectively. The regions indicated in the corresponding light color are false positives.

ing solution. It leverages vehicle-to-everything (V2X) communication to enable agents to share perception information, thereby achieving broader and more comprehensive environmental awareness [4, 17, 30, 40].

Depending on the stage of information transmission, collaborative perception modes can mainly be categorized into early fusion [3], intermediate fusion [2], and late fusion [35]. Current research in collaborative perception predominantly relies on LiDAR [23, 38, 41, 44] or LiDAR-camera fusion [28, 29, 34, 46], with the majority of existing approaches adopting intermediate fusion strategy to achieve a trade-off between detection performance and transmission overhead. However, both modalities are susceptible to environmental degradation, particularly under adverse weather conditions. Under such conditions, sensor inputs undergo various degradations, resulting in missing or false features. Crucially, their shared vulnerability to environmental interference may lead to simultaneous failure in critical scenarios, thereby compromising the robustness of the percep-

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tion system. Despite ongoing efforts to enhance robustness [10, 43], the inherent characteristics of these sensors limit their ability to adapt to the complex degradation distributions present under such conditions.

In recent years, 4D radar has gained increasing attention due to its robust performance across diverse weather scenarios, offering a promising alternative to sensors [1, 24, 26, 37]. Due to their different physical characteristics, LiDAR and 4D radar exhibit complementary strengths. Therefore, fusing LiDAR and 4D radar data is regarded as an effective method to enhance perception reliability under adverse weather conditions. In the context of collaborative perception, LiDAR degradation induced by adverse weather tends to accumulate during data transmission, resulting in a denser and more severe degradation distribution that significantly degrades overall performance. Moreover, LiDAR-4D radar fusion methods still struggle with two issues: 1) *Cross-modal corruption*. Naïve fusion methods [1, 26] may induce mutual corruption between degraded LiDAR and sparse 4D radar, as shown in Fig. 1(a). 2) *Information loss*. Weather-induced degradation already compromises informative content, and denoising-based methods [11, 12] may further discard the remaining useful features while suppressing noise, as shown in Fig. 1(b). On the other hand, the existing collaborative LiDAR-4D radar fusion approach [11] treats the sparse 4D radar modality in the same manner as dense LiDAR during information transmission, integrating both into a unified intermediate collaboration pipeline. While this design achieves notable performance improvements, it doubles the communication overhead. More importantly, applying intermediate collaboration strategy to 4D radar is inefficient, as its inherently sparse measurements incur limited additional cost at the information transmission stage.

To address these challenges, we propose a novel hybrid robust collaborative perception framework (HRCP), designed to improve the collaboration performance under adverse weather conditions through LiDAR-4D radar fusion. In HRCP, we introduce a hybrid collaboration strategy that considers their distinct physical properties and differently processes them during information transmission. This hybrid design not only preserves the original information, but also avoids transmitting dense 4D radar features, reducing additional communication overhead. Furthermore, rather than treating modality fusion as mere feature aggregation, we formulate it as joint reconstruction of a dense and reliable representation. To ensure feature reliability and cross-modal consistency, and to enable comprehensive feature refinement, we propose an adaptive feature enhancement (AFE) module followed by a bidirectional cross-modal gating (BCMG) module to fuse LiDAR and 4D radar features, as shown in Fig. 1(c). Specifically, the BCMG module enables LiDAR and 4D radar to mutually validate

feature reliability and ensure consistent cross-modal representation, and the AFE module enables comprehensive refinement of degraded and suppressed regions to mitigate information loss. To validate the effectiveness of our method, we conduct experiments on both simulation and real-world datasets, including V2X-R [11] and V2X-Radar-C [39].

The main contributions can be summarized as follows:

- We propose a hybrid robust collaborative perception framework, designed to improve the collaboration performance under adverse weather conditions through LiDAR-4D radar fusion. Rather than treating LiDAR-4D radar fusion under adverse weather conditions as mere feature aggregation, we formulate it as joint reconstruction of a dense and reliable representation.
- We introduce a BCMG module that enables LiDAR and 4D radar to mutually validate feature reliability and ensures consistent cross-modal representation following an AFE module that comprehensively refines degraded and suppressed regions to mitigate information loss.
- Extensive experimental results on both simulation and real-world datasets demonstrate that our method outperforms previous state-of-the-art approaches under adverse weather conditions.

2. Related Work

2.1. Collaborative Perception

Collaborative perception enables multiple agents to share complementary perception information, thereby achieving a more comprehensive understanding of the environment. The shared information can be classified into raw sensor data, intermediate features and detection results. Intermediate collaboration, sharing features among agents, has gained increasing attention for its optimal balance between performance and transmission bandwidth. One line of research focuses on enhancing the performance of multi-agent fusion [16, 33, 36]. These approaches improve perception accuracy by designing advanced architectures that better model the cross-agent spatial interactions. Another line of research emphasizes communication-efficient collaboration [4, 8, 9, 25]. These approaches reduce communication overhead by selectively transmitting only the most informative regions or by encoding features into compact yet expressive representations, thereby achieving a favorable balance between perception quality and communication overhead. However, most intermediate collaboration approaches are built upon LiDAR or LiDAR-camera frameworks, and the inherent sensitivity of these sensors to adverse weather conditions still poses significant challenges for collaborative perception.

Despite a single collaboration strategy, some approaches explore a hybrid collaboration strategy. DiscoNet [19] introduces a knowledge extraction framework that utilizes

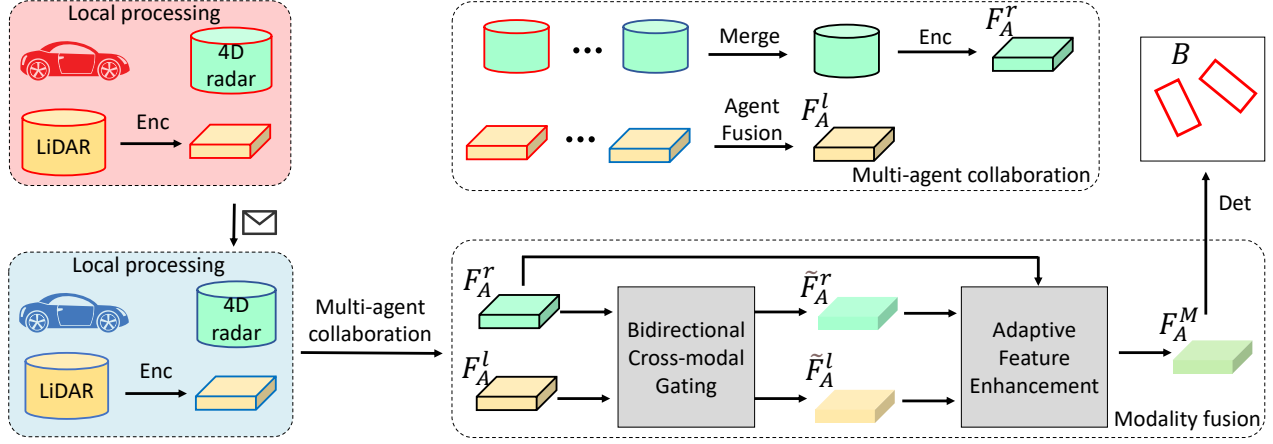


Figure 2. Overview of our proposed HRCP framework for weather-robust collaborative perception. The pipeline consists of four stages: 1) Local processing. 2) Multi-agent collaboration. 3) Modality fusion. 4) Box prediction. In the multi-agent collaboration stage, we employ an early collaboration strategy for 4D radar modality to reduce the additional overhead by incorporating 4D radar. In the modality fusion stage, we introduce a BCMG module and introduce an AFE module to fuse LiDAR and 4D radar features.

early collaboration knowledge to guide the training of the intermediate collaboration model. CoAlign [21] introduces a novel agent-object pose graph modeling to enhance pose consistency among collaborating agents and employs intermediate and late collaboration strategies to aggregate intermediate features. CoSDH [30] introduces a supply-demand awareness model between agents and incorporates late collaboration as compensation for intermediate collaboration. Nevertheless, these approaches remain vulnerable to adverse weather conditions, and their hybrid collaboration strategies are designed for single-modality collaboration.

2.2. 3D Object Detection under Adverse Weather Conditions

Widely used sensors such as LiDAR and cameras may undergo degradation under adverse weather conditions. Physics-based weather simulations [6, 13] have been explored to alleviate data scarcity. Several works [7, 10, 27, 31] propose unsupervised or semi-supervised domain adaptation and knowledge distillation frameworks to adapt models trained under clear-weather conditions to achieve robust performance in adverse weather scenarios. DSRC [43] mitigates the impact of adverse weather conditions by learning density-insensitive and semantic-aware representations in a multi-agent setting. Due to the potential degradation of sensors under adverse weather conditions, some research has explored radar-enhanced object detection. Inter-Fusion [26] leverages a self-attention mechanism to identify cross-modal relations between 4D radar and LiDAR features and aggregates their information. 3D-LRF [1] proposes leveraging the respective advantages of LiDAR and 4D radar in 3D space by introducing a learnable gating

mechanism. L4DR [12] proposes enhancing adaptability to diverse weather conditions by jointly leveraging cross-modal and intra-modal features from LiDAR and 4D radar data through a multi-scale gated fusion strategy. However, these single-agent methods face the cross-modal corruption and information loss issues, and may face significant challenges when extended to multi-agent settings.

3. Method

In this work, we aim to achieve robust collaborative perception under adverse weather conditions through LiDAR-4D radar fusion. In sec. 3.1, we introduce the hybrid collaboration framework for multi-modal data transmission and analyze the strengths of the early collaboration in 4D radar data. Furthermore, we formulate the LiDAR-4D radar fusion as joint reconstruction of a dense and reliable representation, and introduce a BCMG module and an AFE module to conduct it in sec. 3.2 and sec. 3.3, respectively.

3.1. Hybrid Collaboration Framework

We consider a multi-agent setting consisting of N agents collaborating within the scenario, where each agent is equipped with both LiDAR and 4D radar sensors. Let X_i^k denote the observation of modality $k \in \{l, r\}$ of the i -th agent, where l and r denote the LiDAR and 4D radar modalities. As shown in Fig. 2, the collaborative perception process consists of four stages:

1) Local processing. Each agent employs a shared encoder network ϕ_{Enc} to process its local LiDAR observation to obtain LiDAR feature:

$$F_i^l = \phi_{Enc}(X_i^l)_{i=1,2,\dots,N}. \quad (1)$$

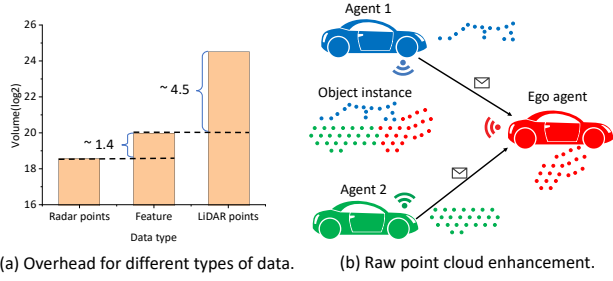


Figure 3. (a) Overhead for different types of data. Transmitting intermediate feature (non-zero elements only) costs $2^{1.4} \times$ more than raw 4D radar point clouds, but $2^{4.5} \times$ less than raw LiDAR point clouds. (b) Sharing raw 4D radar point clouds helps achieve finer-grained perception via collaboration.

2) Multi-agent collaboration. Let the i -th agent be the ego agent. Following local processing, other agents transmit their observation information to the ego agent. For 4D radar modality, we employ early collaboration based on a spatial warping and merging function f_{merge} , and process the fused 4D radar point clouds to generate a multi-agent radar feature F_A^r :

$$F_A^r = \phi_{Enc}(f_{merge}(X_i^r, \{X_{j \rightarrow i}^r\})), j \in \mathcal{N}_i, \quad (2)$$

where $\mathcal{N}_i$ is the set of collaborators of the i -th agent, and $\{X_{j \rightarrow i}^r\}$ is the observation of 4D radar transmitted from agent j to agent i . For LiDAR modality, we employ intermediate collaboration based on an agent fusion network ϕ_A to generate a multi-agent LiDAR feature F_A^l :

$$F_A^l = \phi_A(F_i^l, \{F_{j \rightarrow i}^l\}), j \in \mathcal{N}_i, \quad (3)$$

where $\{F_{j \rightarrow i}^l\}$ is the local LiDAR feature transmitted from agent j to agent i .

3) Modality fusion. A modal fusion network ϕ_M fuses the LiDAR feature with the 4D radar feature to generate a multi-agent multi-modal representation F_A^M :

$$F_A^M = \phi_M(F_A^l, F_A^r). \quad (4)$$

4) Box prediction. Finally, we utilize a detection head to predict the bounding box B from the fused multi-agent multi-modal feature F_A^M .

The insight of our framework lies in a hybrid modality-specific collaboration strategy, which is designed to fit the distinct characteristics of LiDAR and 4D radar sensors in collaborative perception systems. Specifically, LiDAR sensors provide high-resolution, dense point clouds, resulting in a substantial volume of raw data. Directly transmitting such raw data for early collaboration would incur prohibitively high communication bandwidth costs. In contrast, intermediate collaboration, where agents share encoded features, achieves a favorable trade-off between perception accuracy and communication efficiency. This is

why the majority of existing LiDAR-based collaborative perception methods adopt intermediate collaboration. However, the situation differs significantly for 4D radar. Unlike LiDAR, 4D radar sensors inherently generate sparse point clouds with a significantly lower data volume. This low data volume makes early collaboration technically feasible, as shown in Fig. 3(a). More importantly, adopting intermediate collaboration for sparse 4D radar data not only doubles the transmission cost, but may also constrain the performance due to potential information loss during feature encoding. Specifically, the encoding process, which typically involves discretization operations such as voxelization, sacrifices details in comparison to the raw point clouds [4]. While aggregating BEV representations from different agents can enhance the response of potential object regions, the precision may be constrained by incomplete object structure modeling. Raw point clouds preserve the complete object information in the original coordinate space, enabling point-wise alignment and complementary information enhancement across different agents, as shown in Fig. 3(b). Therefore, we adopt early collaboration in the 4D radar modality.

3.2. Modality Fusion Formulation

Following the hybrid collaboration stage, we obtain two distinct multi-agent BEV feature maps from both modalities. These modalities exhibit distinct sensing characteristics under adverse weather conditions: LiDAR provides dense details but suffers from weather-induced degradation, while 4D radar offers reliable sensing but yields extremely sparse measurements. Inspired by this duality, we conduct LiDAR-4D radar fusion under adverse weather conditions from a joint reconstruction perspective. Specifically, we posit the existence of a latent ideal feature $\mathcal{F}$ obtained under ideal conditions. The observed BEV features from each modality are then modeled as distinct degraded views of $\mathcal{F}$:

$$F_A^l = \mathcal{D}_n(\mathcal{F}), F_A^r = \mathcal{D}_s(\mathcal{F}), \quad (5)$$

where $\mathcal{D}_n$ denotes degradation induced by adverse weather, and $\mathcal{D}_s$ represents sparsity inherent to 4D radar sensing.

Under this formulation, the goal of modality fusion is no longer simple aggregation (e.g., concatenation), but joint construction of a dense and reliable representation F_A^M that approximates the latent ideal feature $\mathcal{F}$:

$$F_A^M = \phi_M(F_A^l, F_A^r) \approx \mathcal{F}. \quad (6)$$

Since jointly suppressing dense noise and extracting meaningful cues pose conflicting objectives for a single network, we decompose this problem into a sequential process that first validates feature reliability and ensures cross-modal consistency, then conducts comprehensive refinement.

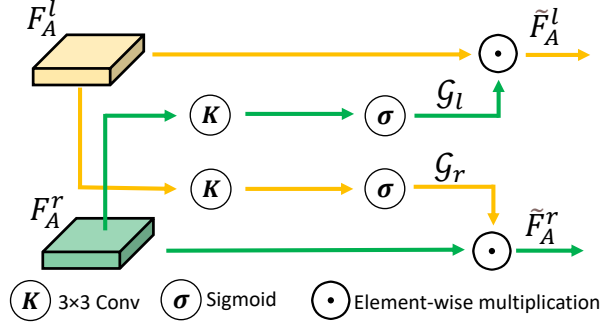


Figure 4. The bidirectional cross-modal gating module. It adaptively generates gating weights based on each modality and applies these weights to validate the other modality through element-wise multiplication.

3.3. Multi-modal Fusion

In the multi-modal context, each modality serves as a validator for the other. Leveraging the weather-robust nature of 4D radar features, we identify spatial regions in the LiDAR feature map that remain reliable under adverse conditions, distinguishing them from those severely degraded. Conversely, the dense structural details provided by LiDAR help distinguish meaningful returns associated with actual objects from non-structural clutter.

To exploit the complementary strengths of LiDAR and 4D radar, we introduce a Bidirectional Cross-Modal Gating (BCMG) module that enables feature reliability and ensures cross-modal consistency in a mutual manner. Specifically, for each modality, we generate a soft attention gate conditioned on the other modality:

$$G_l = \sigma(W_{r \rightarrow l} * F_A^r), G_r = \sigma(W_{l \rightarrow r} * F_A^l), \quad (7)$$

where G_l and G_r are gating maps for the LiDAR feature and 4D radar feature, respectively, $W_{r \rightarrow l}$ and $W_{l \rightarrow r}$ are learnable convolutional kernels, $*$ denotes convolution operation, and σ is the sigmoid activation function. Subsequently, these gates are applied to the original modality features via element-wise multiplication:

$$\tilde{F}_A^l = G_l \odot F_A^l, \tilde{F}_A^r = G_r \odot F_A^r. \quad (8)$$

Figure 4 shows the BCMG module. Therefore, we obtain a radar-verified LiDAR feature $\tilde{F}_A^l$ where unreliable, weather-degraded regions are suppressed, and a LiDAR-guided radar feature $\tilde{F}_A^r$ where non-structural responses are suppressed.

Though the BCMG improves the reliability of modality-specific features, it is fundamentally a subtractive process. Moreover, weather-induced degradation is not only spurious noise, but also signal attenuation, as shown in Fig. 5. Only relying on these suppressed representations is prone to suboptimal performance.

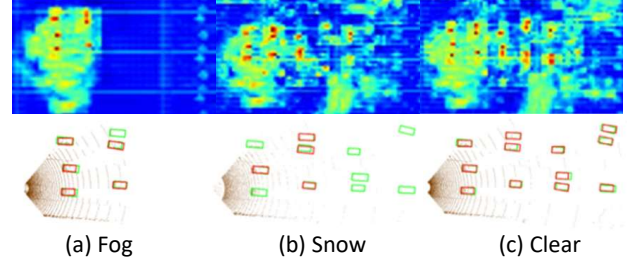


Figure 5. Different weather-induced degradations in LiDAR sensors. The degradation involves not only the introduction of spurious noise but also signal attenuation.

To overcome this limitation, we introduce an adaptive feature enhancement (AFE) module to refine the degraded and suppressed regions through a residual network with sparse channel-wise attention, thereby enhancing the overall feature quality. Specifically, we concatenate the radar-verified LiDAR feature $\tilde{F}_A^l$, the LiDAR-guided radar feature $\tilde{F}_A^r$, and the original 4D radar feature F_A^r to preserve both consistent context and raw cues. The concatenated representation is then projected into a shared latent space via a learnable embedding function $f_{\text{emb}}(\cdot)$:

$$F_{\text{emb}} = f_{\text{emb}}(\text{Cat}[\tilde{F}_A^l, \tilde{F}_A^r, F_A^r]). \quad (9)$$

To adaptively calibrate channel responses and identify spatial regions need to be refined, we apply sparse channel attention:

$$s = \sigma(\text{Conv1D}(\text{GMP}(F_{\text{emb}}))), \quad (10)$$

where $\text{GMP}(\cdot)$ is the global max pooling operation. The attention-modulated feature is obtained through a skip connection to enhance discriminative channels while preserving base information:

$$F_{\text{att}} = F_{\text{emb}} + (F_{\text{emb}} \odot s). \quad (11)$$

Finally, an output projection f_{out} maps F_{att} to a residual term that further corrects the radar-verified LiDAR feature:

$$F_A^M = \tilde{F}_A^l + f_{\text{out}}(F_{\text{att}}). \quad (12)$$

Figure 6 shows the AFE module. By leveraging sparse channel-wise attention, AFE adaptively focuses on regions carrying highly informative signals. The following residual design not only refines the degraded and suppressed representation under adverse weather conditions, but also learns the latent interactions between both modalities to enhance feature representation under clear conditions.

3.4. Loss Function

Since the ideal feature is unavailable even under clear weather conditions in practice, we adopt an end-to-end

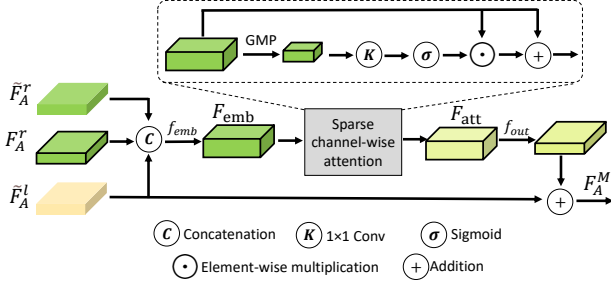


Figure 6. The adaptive feature enhancement module. It applies sparse channel-wise attention and generates a residual correct term to capture interactions between both modalities and refine the degraded and suppressed regions.

learning strategy without supervision on latent features. In the downstream detection task, the detection head is composed of a classification head and a regression head, and they extract the classification and regression output, respectively, to predict the bounding box. Each head consists of a single convolutional block. We leverage the focal loss [20] for classification (denoted as $\mathcal{L}_{cls}$) and the smooth absolute error loss for bounding box regression (denoted as $\mathcal{L}_{reg}$). The total loss is:

$$\mathcal{L}_{total} = \beta_{cls}\mathcal{L}_{cls} + \beta_{reg}\mathcal{L}_{reg}, \quad (13)$$

where β_{cls}, β_{reg} are hyperparameters.

4. Experiment

4.1. Dataset and Metric

We evaluate HRCF on V2X-R [11] and V2X-Radar-C [39]. V2X-R is a simulated V2X collaborative perception dataset, jointly developed using CARLA [5] and OpenCDA [32] platforms. It contains LiDAR and 4D radar point clouds under simulated adverse weather conditions. V2X-Radar-C is a large-scale real-world multi-modal dataset featuring 4D radar for collaborative perception. Performance is measured by mean Average Precision (mAP) at Intersection-over-Union (IoU) thresholds of 0.3, 0.5, and 0.7.

4.2. Compared Methods

The implementation of the above collaboration framework relies on the agent fusion and modal fusion networks. We follow the implementations of MDD [11] to extend prior works to collaborative LiDAR-4D radar fusion. Multi-agent single-modal methods such as AttFuse [34], V2X-ViT [33], CoBEVT [36] and CoAlign [21] are adapted by treating 4D radar like LiDAR and concatenating the features from both modalities. For multi-modal methods such as InterFusion [26] and L4DR [12], we derive their multi-agent versions by incorporating a self-attention-based agent-fusion

module. Our framework adopts the above extended methods for multi-agent and multi-modal fusion for fairness.

4.3. Implementation Details

Following the setup in V2X-R [11], we select one vehicle as the ego agent and evaluate detection performance in the vicinity of the ego agent, within a spatial range of $x \in [0, 140]$ and $y \in [-40, 40]$. We assume that all agents have a communication range of 70m. All the agents outside this broadcasting radius of the ego agent do not participate in any collaboration. In addition, all of our evaluation results are within the camera field of view of the ego agent. All methods employ PointPillar [15] as the backbone to extract features from the point clouds for fairness. It converts point cloud data into voxels with a resolution of 0.4m for both height and width. We implement the proposed and comparative models using the PyTorch framework [22] and train them on eight RTX 3090 24G GPUs using the Adam optimizer [14] with $lr = 1e-3, \beta_1 = 0.9, \beta_2 = 0.999$. The hyperparameters $\beta_{cls} = 1, \beta_{reg} = 2$. To simulate the pose error, we add Gaussian noise with a standard deviation of σ_t for localization errors and σ_r for heading errors.

4.4. Quantitative Evaluation

Performance comparison under different simulated weather conditions on V2X-R dataset. Table 1 presents the 3D object detection performance of our HRCF in comparison with several state-of-the-art methods under clear, fog and snow conditions. Under fog conditions, scattering and signal attenuation induce dense false points at close range and blur distant object boundaries, while under snow conditions, snowflake occlusion reduces valid surface points and increases false return density. Among well-designed multi-agent collaboration methods, CoAlign achieves better performance than others under adverse weather conditions, due to the agent-object pose graph modeling for pose errors, which can be treated as another form of degradation. Nevertheless, our HRCF still achieves superior performance, with improvements of 4.42% and 9.09% in mAP@0.7 under fog and snow conditions, respectively, indicating the adaptability of our framework in multi-agent collaboration scenarios. Compared to the well-designed LiDAR-4D radar fusion method L4DR, HRCF achieves performance gains of 5.10% and 8.70% in mAP@0.7 under fog and snow conditions, respectively, indicating the effectiveness of our framework in multi-agent collaboration scenarios. Notably, our method also outperforms all baseline methods under clear weather conditions, demonstrating that our modality fusion formulation promotes more effective latent interactions between modalities and achieves fusion performance closer to the ideal case. Particularly, HRCF achieves a significant performance improvement in the mAP@0.7 metric, validating its advan-

Table 1. 3D detection performance comparison under clear, fog and snow conditions.

Method	Clear			Fog			Snow		
	mAP@0.3	mAP@0.5	mAP@0.7	mAP@0.3	mAP@0.5	mAP@0.7	mAP@0.3	mAP@0.5	mAP@0.7
AttFuse	94.03	92.69	82.45	85.48	80.60	56.76	87.87	84.41	63.54
V2X-ViT	91.82	89.73	72.75	86.23	81.17	52.43	87.76	84.81	57.06
CoBEVT	87.71	87.21	74.61	82.14	79.25	55.86	81.22	80.27	61.93
CoAlign	91.41	90.32	79.37	85.94	81.33	61.62	88.10	85.22	67.54
InterFusion	78.66	75.21	43.80	76.14	70.50	36.49	79.55	75.19	40.53
L4DR	94.32	92.92	80.15	85.85	81.26	60.94	86.78	84.45	67.93
MDD	92.13	90.26	79.21	84.86	81.53	59.52	88.07	85.24	63.31
Ours	95.10	94.01	84.07	86.96	82.86	66.04	92.34	90.54	76.63

Table 2. Detection mAP@0.5 with localization error and heading error under clear and fog conditions.

Conditions	Clear					Fog				
$\sigma_r/\sigma_t(m/^{\circ})$	0.1/0.1	0.2/0.2	0.3/0.3	0.4/0.4	0.5/0.5	0.1/0.1	0.2/0.2	0.3/0.3	0.4/0.4	0.5/0.5
AttFuse	91.93	91.12	90.17	88.55	85.79	80.19	79.07	78.97	77.61	75.03
V2X-ViT	90.04	88.88	86.27	83.62	79.37	80.90	80.57	79.68	77.49	75.13
CoBEVT	87.23	86.91	86.07	83.89	81.48	77.80	76.84	74.47	72.00	68.21
CoAlign	90.20	89.51	88.32	86.64	83.82	80.89	80.88	79.98	78.99	76.34
InterFusion	73.05	72.08	69.51	66.19	62.66	69.83	68.40	65.09	61.64	59.08
L4DR	92.44	91.45	89.79	87.28	85.64	81.57	81.05	79.80	78.62	75.35
MDD	90.40	89.43	88.25	87.36	85.42	80.78	78.27	75.60	73.82	71.97
Ours	93.83	92.99	92.30	91.18	90.15	82.67	81.79	81.47	80.59	79.78

tages in high-precision collaborative perception tasks.

Pose robustness analysis on V2X-R dataset. To evaluate the sensitivity of existing methods to localization and heading errors, we conduct extensive experiments under clear and fog conditions. As shown in Table 2, the detection accuracy of all methods rapidly decreases as the localization and heading errors increase. Under both conditions, our HRCP consistently achieves the best performance across all evaluated error levels. Notably, as these errors increase, our HRCP maintains the smallest decline in detection accuracy among all compared methods.

Table 3. 3D detection performance on V2X-Radar-C.

	mAP@0.3	mAP@0.5	mAP@0.7
AttFuse	44.11	38.58	25.90
V2X-ViT	49.80	44.51	27.11
CoBEVT	49.15	<u>46.04</u>	26.44
CoAlign	<u>52.55</u>	43.91	29.66
InterFusion	40.57	32.81	12.47
L4DR	48.06	42.33	<u>30.71</u>
MDD	34.81	28.53	18.19
Ours	54.08	49.72	33.19

Performance comparison on real-world V2X-Radar-C dataset. We further conduct experiments on the real-world V2X-Radar-C dataset. As shown in Table 3, our HRCP

framework still perform well in the real-world scenario, achieving the best results across all metrics. Specifically, it outperforms the second-best results by 1.53%, 3.68%, and 2.48% on mAP@0.3, mAP@0.5, and mAP@0.7, respectively. These results confirm the effectiveness of our design under real-world conditions.

4.5. Ablation Study

To validate the necessity of LiDAR-4D radar fusion for collaborative perception and to assess the contribution of each component, we conduct a comprehensive ablation study under adverse weather conditions, as shown in Table 4.

Ablation on modalities. The radar-only method achieves moderate detection accuracy, particularly on mAP@0.7. Though it is robust to weather (not weather-degraded), the inherent sparsity prevents it from achieving high-precision detection, making it unsuitable as a standalone detector. In contrast, LiDAR is commonly used alone, but its effectiveness is severely compromised under adverse weather conditions due to the induced degradation. The significant performance improvement achieved by fusing LiDAR and 4D radar highlights the effectiveness of multi-modal integration. This synergy makes multi-modal collaborative perception a highly viable solution for maintaining effectiveness under adverse weather conditions.

Ablation on proposed components. The rows in (d), (e), and (f) of Table 4 illustrate the impact of each proposed

Table 4. Ablation study on the modalities and the proposed components.

Method	Modality		Components		Fog	Snow
	R	L	BCMG	AFE	mAP@0.3/0.5/0.7	mAP@0.3/0.5/0.7
(a) Radar Only	✓				82.04/74.25/42.53	82.04/74.25/42.53
(b) LiDAR Only		✓			67.26/65.86/50.28	82.23/78.24/51.75
(c) LiDAR-radar fusion	✓	✓			85.48/80.60/56.76	87.87/84.41/63.54
(d) Ours without AFE	✓	✓	✓		87.21 /82.30/59.91	91.42/88.62/64.61
(e) Ours without BCMG	✓	✓		✓	84.15/78.70/60.05	91.41/88.78/70.87
(f) Ours	✓	✓	✓	✓	86.96/ 82.86/66.04	92.34/90.54/76.63

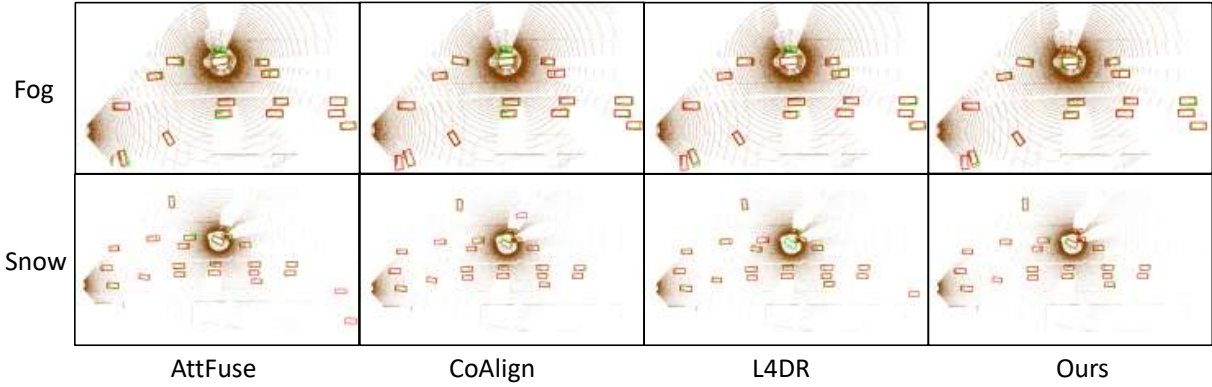


Figure 7. Visualization of AttFuse, CoAlign, L4DR and HRCP under fog and snow conditions. Green and red boxes denote ground truth and predictions, respectively. We highlight the obvious errors in blue bounding boxes.

component. When BCMG is incorporated, each modality serves as a validator for the other. Through cross-modal mutual validation, both modalities benefit from enhanced feature consistency and reliability, thereby leading to improved overall performance. In (e), although we observe only marginal improvements or even a slight drop compared to (d) in mAP@0.3 and mAP@0.5, there is a significant benefit in mAP@0.7. This improvement stems from the sparse channel-wise attention, which adaptively calibrates channel-wise responses to refine the degraded regions. (f) presents our full model, which integrates the two components. By ensuring feature reliability and cross-modal consistency, and simultaneously leveraging sparse channel-wise responses to selectively refine degraded and suppressed regions, our approach effectively combines the strengths of both components for superior performance.

4.6. Qualitative Evaluation

Figure 7 presents the collaborative detection results of AttFuse, CoAlign, L4DR, and our HRCP under fog and snow conditions. We see that our HRCP produces more accurate predictions across these scenarios, demonstrating superior performance under weather-induced degradation. The reason is that our method ensures feature reliability and cross-

modal consistency, and calibrates channel-wise feature responses to refine degraded and suppressed regions.

5. Conclusion

In this paper, we propose HRCP, designed to improve the collaboration performance under adverse weather conditions through LiDAR-4D radar fusion. To reduce the additional overhead, we adopt early collaboration strategy in 4D radar modality, while retaining intermediate collaboration strategy in LiDAR modality. Additionally, we propose a BCMG module to ensure feature reliability and cross-modal consistency, and an AFE module to comprehensively refine degraded and suppressed regions. To validate the effectiveness of our method, we conduct experiments on both simulation dataset V2X-R and real-world dataset V2X-Radar-C. Extensive experimental results demonstrate that our method outperforms previous state-of-the-art approaches under adverse weather conditions.

Limitations and future work. Incorporating 4D radar into the collaboration pipeline significantly improves perception under adverse weather conditions, but it inevitably adds communication overhead. In future work, we will explore a new joint representation to achieve the trade-off between perception accuracy and communication overhead.

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